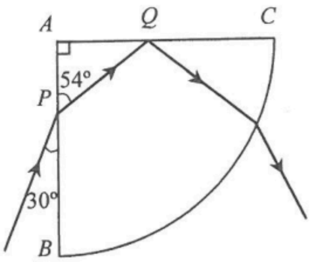
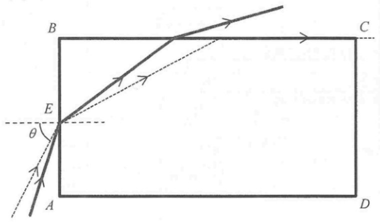
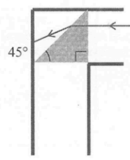
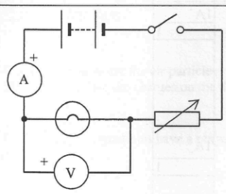
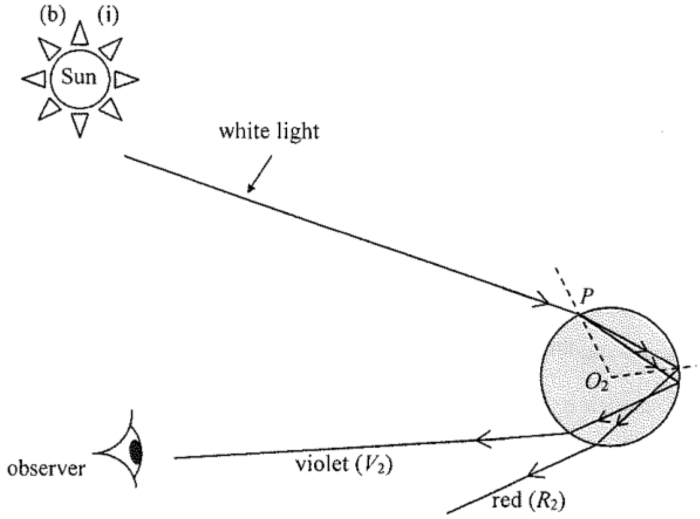
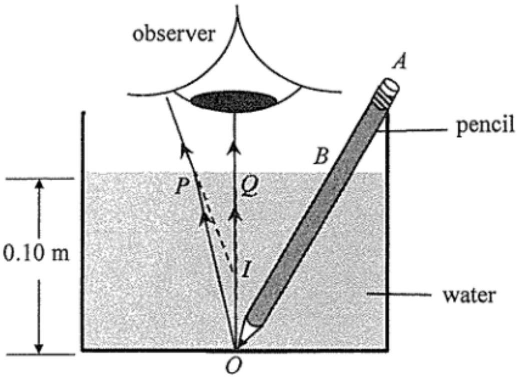
Solution	Marks	Remarks
5. (a) $n = \dfrac{\sin i}{\sin r}$ (or $n_1 \sin\theta_1 = n_2 \sin\theta_2$)	1M	
$= \dfrac{\sin(90^\circ - 30^\circ)}{\sin(90^\circ - 54^\circ)} = \dfrac{\sin 60^\circ}{\sin 36^\circ}$		
$= 1.47$	1A	Accept 1.47 to 1.50
	2	
(b) $\sin c = \dfrac{1}{n} = \dfrac{1}{1.47}$	1M	
$c = 42.7^\circ$ (for $n = 1.50$, $c = 41.8^\circ$) Because incident angle at face AC (= 54°) > c (= 42.7°).	1M	
	2	
(c) 		
Reflected ray correct, $i = r$	1A	
Emergent ray away from normal.	1A	
	2	
(d) A spectrum is seen.	1A	
Or splitting into different colours occurs.	1A	
	1	

7. (a) (i) At the critical angle c,		
$\dfrac{\sin 90^\circ}{\sin c} = n$		
$\dfrac{1}{\sin c} = 1.36$	1M	
$c = 47.3^\circ$	1A	
	2	
(ii) Angle of refraction at $E = 90^\circ - 47.33^\circ = 42.67^\circ$	1M	
By Snell's law		
$\dfrac{\sin\theta}{\sin 42.67^\circ} = 1.36$	1M	
$\theta = 67.2^\circ$	1A	
	3	
(iii) 	2A	
	2	
(b) (i) 	1A	
The angle of incidence of the light ray from the object is (45°, which is) less than the critical angle of the plastic prism.	1A	
Total internal reflection will not occur and the image may not be clear enough for observation.	1A	
	3	
(ii) Glass prism (with critical angle less than 45°)	1A	
Or		
Plane mirror	1A	
	1	

Solution	Marks	Remarks
8. (a) 	1A	Correct symbols of light bulb, variable resistor and voltmeter
	1A	Correct positions
	1A	Correct positive terminal connection for the voltmeter
	3	
(b) As the voltage across the light bulb increases, the temperature of the light bulb increases,	1A	
thus its resistance increases.	1A	
	2	
(c) $R = \dfrac{V}{I}$ is the definition of resistance. It is applicable to all conductors.	1A	
	1	
(d) At $V = 0.1$ V		
$R = \dfrac{V}{I} = \dfrac{0.1}{76\times10^{-3}} = 1.32\ \Omega$	1A	
At $V = 2.5$ V		
$R = \dfrac{V}{I} = \dfrac{2.5}{250\times10^{-3}}$	1A	for reading correct values (ignore the order of magnitude) from the graph
$= 10\ \Omega$	1A	
	3	
(e) $R = \rho\dfrac{l}{A}$		
$l = \dfrac{RA}{\rho}$		
$= \dfrac{1.32\times(1.66\times10^{-9})}{5.6\times10^{-8}}$	1M+1M	
$= 0.039$ m	1A	
	3	

7.	(a)	(i)	Angle of incidence at A, $i_A = 90° - 30° = 60°$	1A	
				1	
		(ii)	$n_g \sin c = n_c \sin 90°$	1M	1M for correct equation with angle of refraction in the cladding = 90°
			$\Rightarrow \dfrac{n_g}{n_c} = \dfrac{1}{\sin c} \geq \dfrac{1}{\sin 60°} = 1.1547005 \approx 1.15$	1A	Accept 1.2 or $\frac{2}{\sqrt{3}}$
				2	
		(iii)	Total internal reflection.	1A	Correct spelling
			For light rays from O with $\theta > 30°$, they will fail to undergo total internal reflection.	1A	Accept $\theta \geq 30°$
				2	
	(b)	(i)	Some light / energy (of the narrow light pulse) taking the shortest path (i.e. OD) arrives first while the rest of the energy taking longer paths arrives later.	1A 1A	1A for 'taking different paths' 1A for 'different arrival time'
			Therefore the pulse is broader and the height of the pulse is lower (smaller intensity).		NOT accept: Some of the energy (going at angles larger than 30° from the axis) would be lost due to leakage, thus lower intensity.
				2	
		(ii)	The refractive index of cladding should be increased,	1A	
			as $\dfrac{n_g}{n_c} = \dfrac{1}{\sin c}$, by making c / $\sin c$ larger, only those light rays close to the axis / within a smaller θ are totally internal reflected,	1A	0 A for the explanation if the choice for the change of n_c is INCORRECT.
			(such that $\dfrac{n_g}{n_c}$ getting closer to 1).		Note: More light rays will escape the optical fibre for a larger n_c.
				2	

Solution	Marks	Remarks
6. (a) (i) $\sin r = \frac{\sin 45°}{1.325}$	1M	
$r = 32.253454° \approx 32.3°$	1A	
	2	
(ii) $c = \sin^{-1}\frac{1}{1.325}$	1M	
$c = 49.000646° \approx 49.0°$	1A	
r (or $\angle OBA$) $= 32.3° < c = 49.0°$, therefore total internal reflection does not occur at point B.	1A	
	3	
(b) (i) 	2A	
	2	
(ii) There is more light energy absorbed by water droplet	1A	
due to a longer path.	1A	
Or There is more light energy transmitted to air at the point of	1A	
reflection as the light undergoes reflection twice.	1A	
	2	

8. (a) 	1A 1M	1M for image I formed by the two refracted rays.
	2	
(b) $1.33 = \frac{\sin r}{\sin 1.5°}$ $r = 1.995175° \approx 2.00°$	1M 1A	
	2	
(c) $\frac{h_1}{h} = \frac{\tan 1.5°}{\tan 2.00°} = 0.751681$ (or $\tan 1.5° = \frac{PQ}{0.10}$ & $\tan 2° = \frac{PQ}{h_1}$) $\Rightarrow h_1 \approx 0.075$ m (i.e. 7.5 cm)	1M 1A	e.c.f. from (b), explicit step / method shown Accept: $\frac{\text{real depth}}{\text{apparent depth}} = \frac{h}{h_1} = n$
	2	